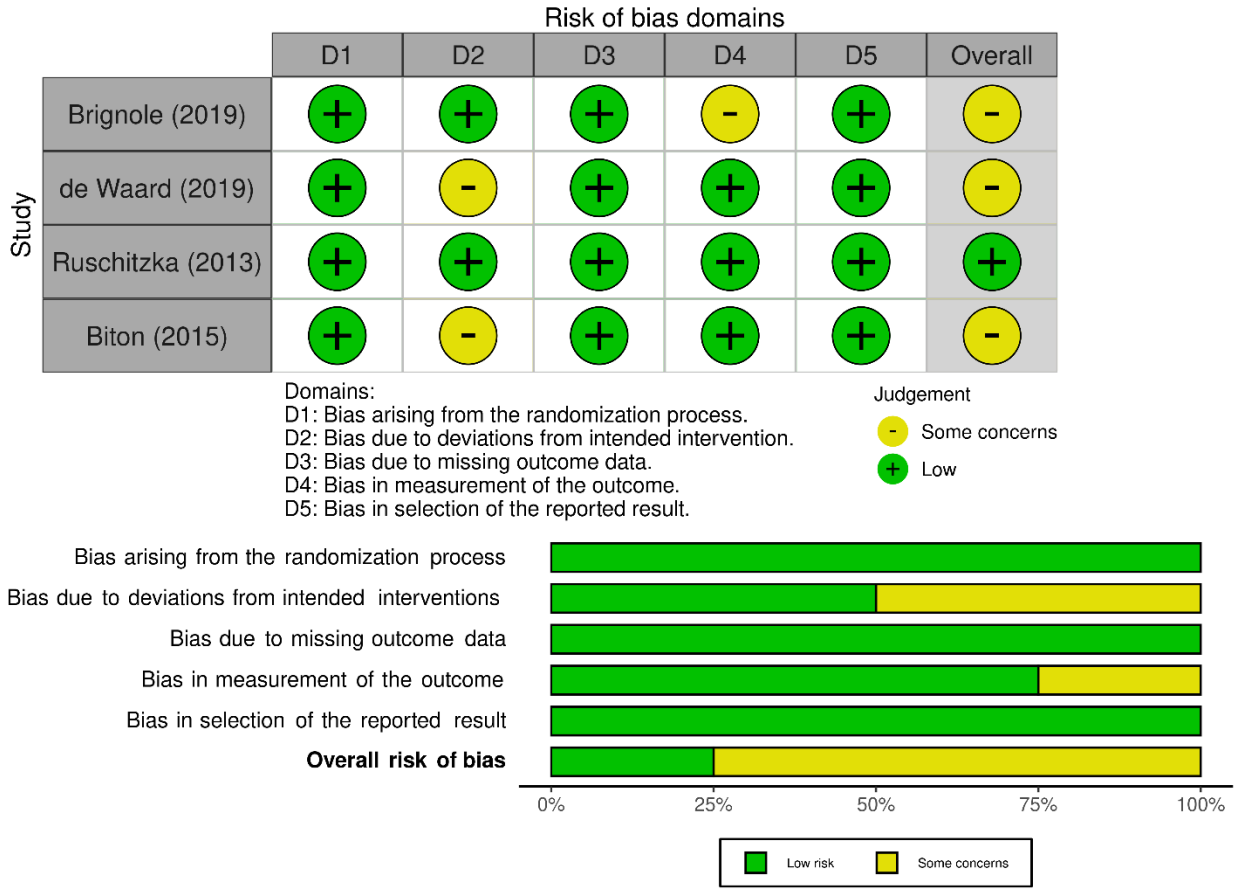
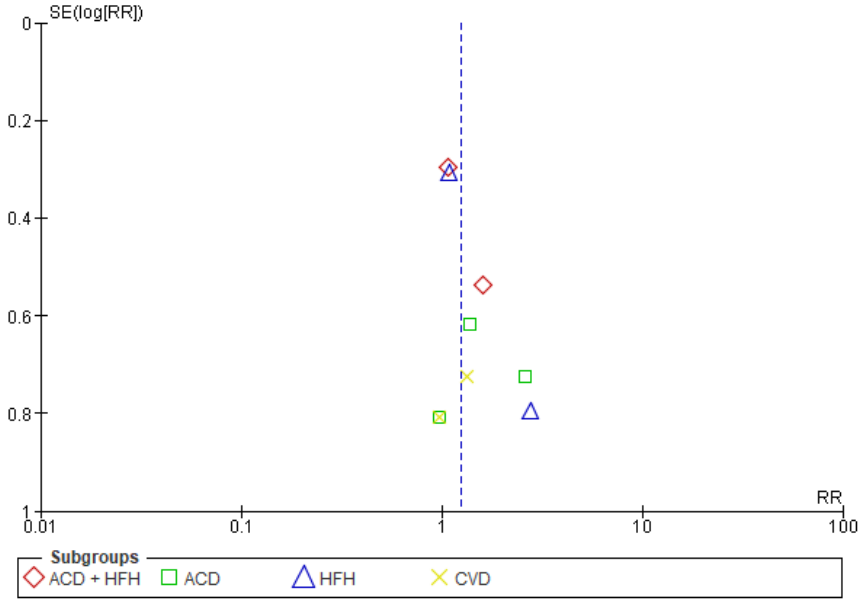
Supplementary Material

Supplemental Figure S1. Risk of Bias Analysis for Included CRT RCTs. Shown are the individual risk of bias domains for each study based on low (green), moderate (yellow), and high (red) study bias (top panel); along with the overall summary of each domain and the study analysis (bottom panel). Two studies, Elanchenny et al. (2012) and Steffel et al. (2016), provide secondary analyses of primary trial data from Biton et al. (2015) and Rushitzka et al. (2013), respectively. **Abbreviations:** CRT = cardiac resynchronization therapy; RCTs = randomized controlled trials.

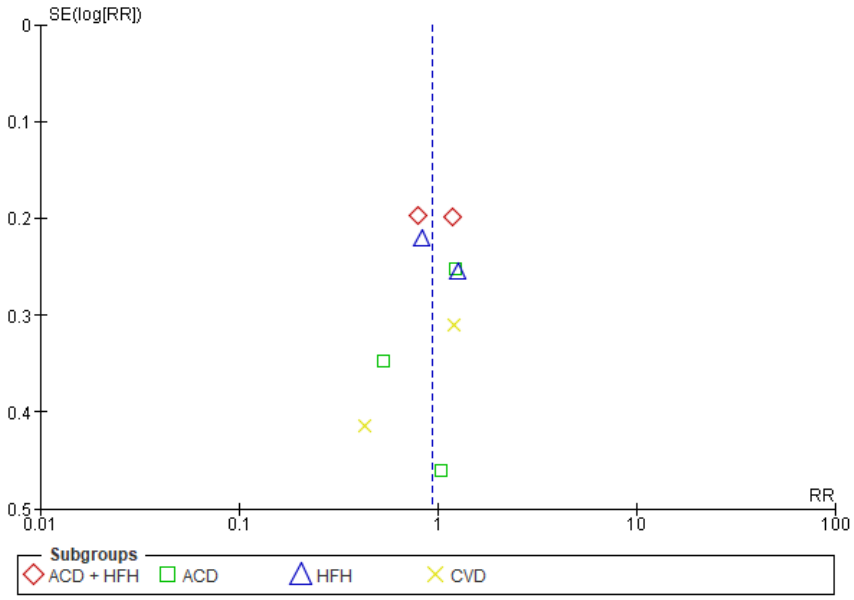


Supplemental Figure S2. Funnel Plots for Detection of Systematic Heterogeneity by Sex in CRT RCTs. Shown are funnel plots for publication bias for sex, specifically male (A) and female (B) patient cardiovascular outcomes and mortality. Funnel plots pertaining to race and age were not included as a limited number of studies reported clinical outcomes such that discerning chance from real asymmetry was likely negligible for meaningful analyses. **Abbreviations:** CRT = cardiac resynchronization therapy; RCTs = randomized controlled trials.

A. Publication Bias in Male Patient Cardiovascular Outcomes and Mortality.

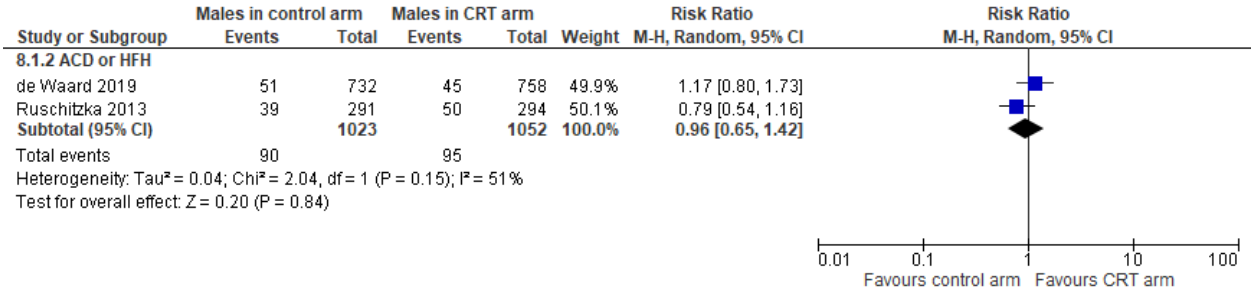


B. Publication Bias in Female Patient Cardiovascular Outcomes and Mortality.

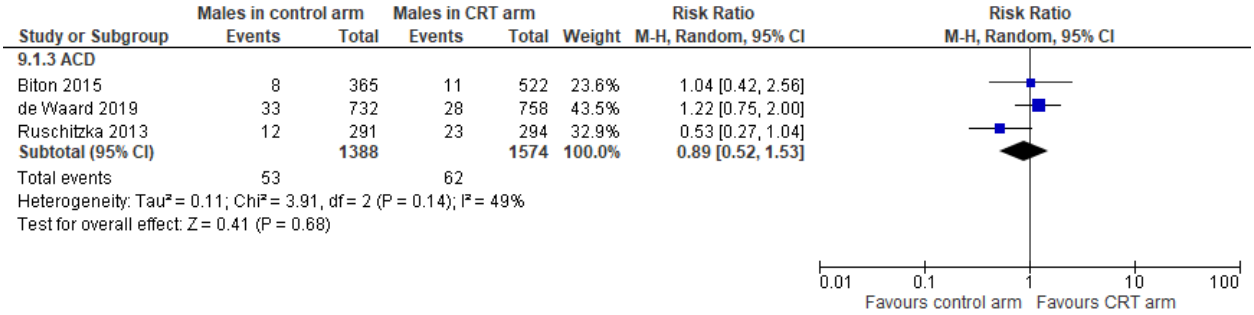


Supplemental Figure S3. Cardiovascular Outcomes and Mortality Between Treatment and Control Arms for Males in CRT RCTs. Shown are Forest Plot analyses for cardiovascular outcomes and mortality in males between CRT and control arms for all-cause death or HFH (A), all-cause death (B), HFH (C), and cardiovascular death (D). **Abbreviations:** ACD = all-cause death; CVD = cardiovascular death; CRT = cardiac resynchronization therapy; HFH = heart failure hospitalization; RCTs = randomized controlled trials.

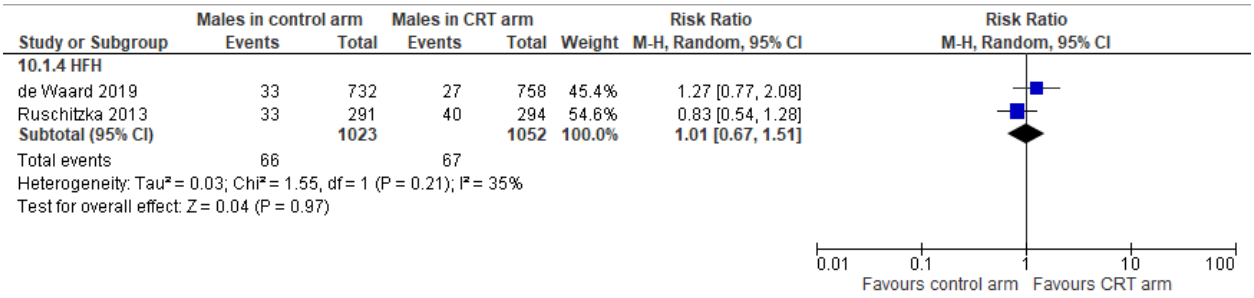
A. Reported All-Cause Death or Heart Failure Hospitalization Between CRT and Control Arms in Males.



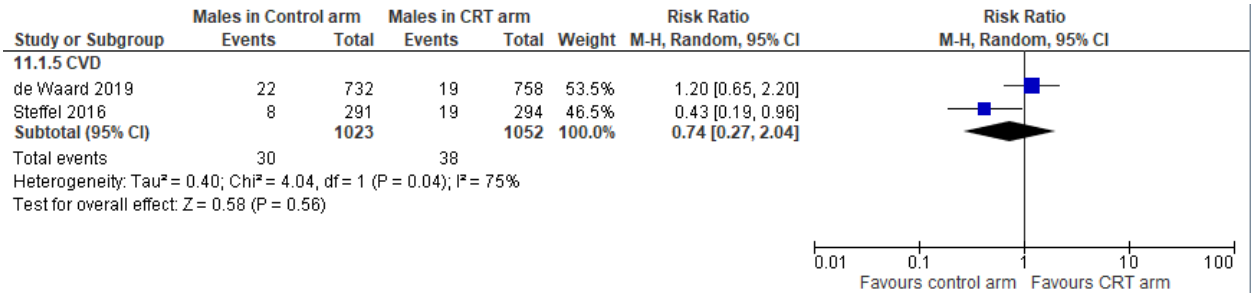
B. Reported All-Cause Death Between CRT and Control Arms in Males.



C. Reported Heart Failure Hospitalization Between CRT and Control Arms in Males.

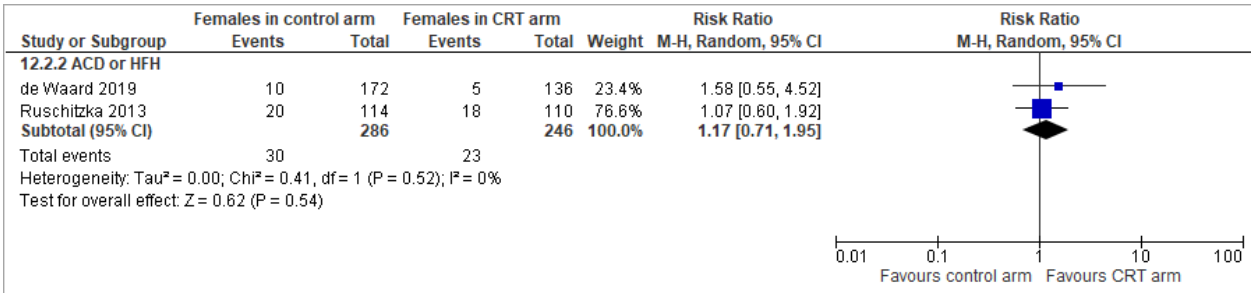


D. Reported Cardiovascular Death Between CRT and Control Arms in Males.

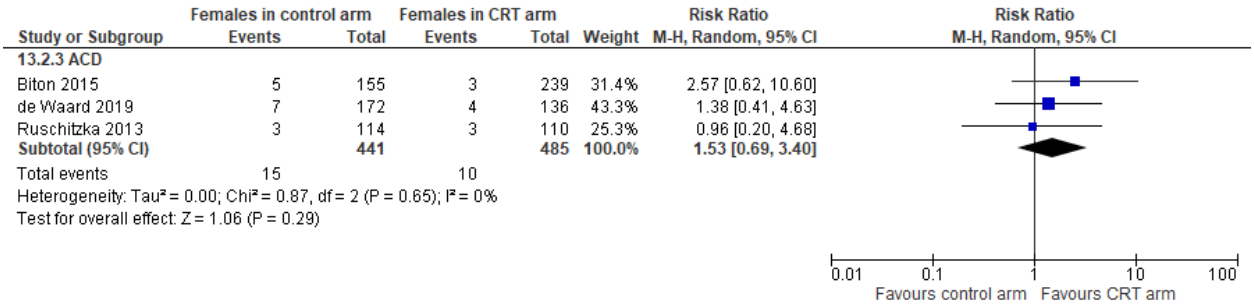


Supplemental Figure S4. Cardiovascular Outcomes and Mortality Between Treatment and Control Arms for Females in CRT RCTs. Shown are Forest Plot analyses for cardiovascular outcomes and mortality in females between CRT and control arms for all-cause death or HFH (A), all-cause death (B), HFH (C), and cardiovascular death (D). **Abbreviations:** ACD = all-cause death; CVD = cardiovascular death; CRT = cardiac resynchronization therapy; HFH = heart failure hospitalization; RCTs = randomized controlled trials.

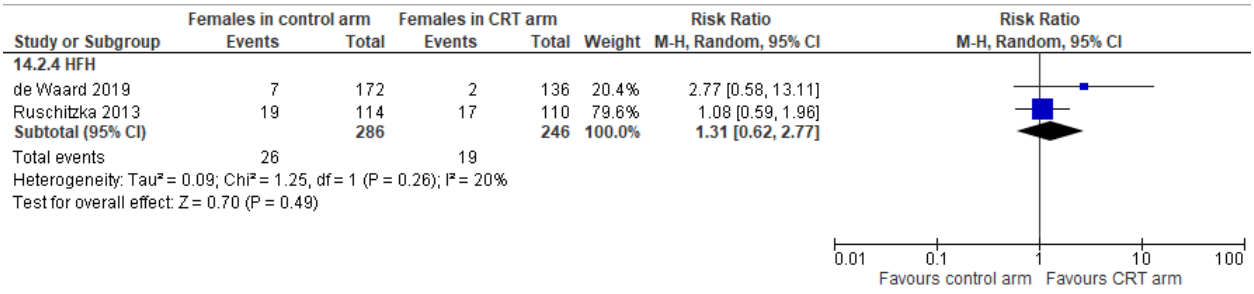
A. Reported All-Cause Death or Heart Failure Hospitalization Between CRT and Control Arms in Females.



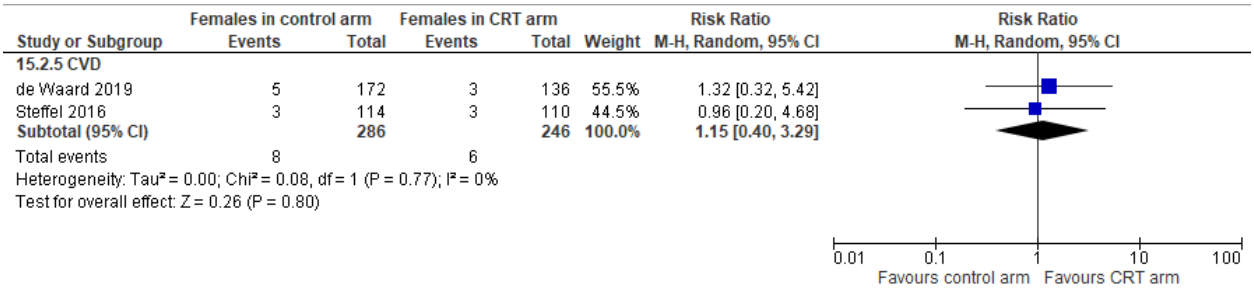
B. Reported All-Cause Death Between CRT and Control Arms in Females.



C. Reported Heart Failure Hospitalization Between CRT and Control Arms in Females.



D. Reported Cardiovascular Death Between CRT and Control Arms in Females.



Supplemental Table S1: Summary of Search Strategy for CRT RCTs.

1	[Population: Cardiovascular Diseases]
2	exp Cardiovascular Diseases/ (2385474)
3	((cardiovascular or angiocardiovascular or angiocardiopath*) adj4 (disease* or disorder* or syndrome*)).tw,kw. (191621)
4	(stroke* or heart failure* or myocardial infarction or MI).tw,kw. (569337)
5	or/2-4 (2615415)
6	[Intervention: CRT]
7	exp Cardiac Pacing, Artificial/ [includes Cardiac Resynchronization Therapy/] (24740)
8	exp Pacemaker, Artificial/ [includes Cardiac Resynchronization Therapy Devices/] (27214)
9	(pacing adj3 (biventricular or atrio-biventricular or artificial or cardiac)).tw,kw. (5255)
10	(cardiac adj3 resynchroni*).tw,kw. (7783)
11	(pacemaker* or CRT-P or CRT or CRT-D).tw,kw. (53157)
12	or/7-11 (79026)
13	[Outcomes: MACE]
14	Death, Sudden, Cardiac/ (15457)
15	exp Mortality/ (382745)
16	exp Hospitalization/ and ((cardiac or heart) adj3 (failure or arrest)).tw,kw. (12665)
17	(outcome* or mortality or adverse event* or cardiovascular event* or death* or hospitali* or complication*).tw,kw. (3661997)
18	(MACE* or major adverse cardiovascular event* or major cardiovascular event*).tw,kw. (19070)
19	or/14-18 (3819821)
20	5 and 12 and 19 (13447)
21	[Study Types]
22	exp randomized controlled trial/ (511808)
23	exp Randomized Controlled Trials as Topic/ (138325)
24	(random* or placebo*).mp. (1448979)
25	(blind or double-blind).mp. (267323)
26	(clinical trial, all or controlled clinical trial or meta analysis or randomized controlled trial).pt,ti. (761365)
27	or/22-26 (1634523)
28	20 and 27 (2079)
29	limit 28 to "humans only (removes records about animals)" (2054)

Note: Sample search strategy is shown above for Ovid (Medline) from 1946 to August 07, 2020. Similar search strategies were carried out for the other electronic databases. **Abbreviations:** CRT = cardiac resynchronization therapy; RCTs = randomized controlled trials.

Supplemental Table S2: Excluded Landmark CRT RCTs.

Study #	Trial Name	Study Reference	Reason for Exclusion
1	CARE-HF	Cleland JG, Daubert JC, Erdmann E, et al. The effect of cardiac resynchronization on morbidity and mortality in heart failure. N Engl J Med. 2005;352(15):1539-1549. doi:10.1056/NEJMoa050496	Unable to distinguish sex and age-based event rates by treatment group (figure 2). No differentiation present between the CRT and control groups for subgroup specified event numbers.
2	COMPANION	Bristow MR, Saxon LA, Boehmer J, et al. Cardiac-resynchronization therapy with or without an implantable defibrillator in advanced chronic heart failure. N Engl J Med. 2004;350(21):2140-2150. doi:10.1056/NEJMoa032423	No outcome event rates reported for sex, age, or race. Absence of meta regression precluded extraction of the hazard ratios reported for sex and age in figure 2 (see study limitations).
3*	MADIT-CRT	Moss AJ, Hall WJ, Cannom DS, et al. Cardiac-resynchronization therapy for the prevention of heart-failure events. *N Engl J Med*. 2009;361(14):1329-1338. doi:10.1056/NEJMoa0906431	Unable to distinguish sex and age-based event rates by treatment group (figure 3). No differentiation present between the CRT and control groups for subgroup specified event numbers.
4*	RAFT	Tang AS, Wells GA, Talajic M, et al. Cardiac-resynchronization therapy for mild-to-moderate heart failure. N Engl J Med. 2010;363(25):2385-2395. doi:10.1056/NEJMoa1009540	Unable to distinguish sex and age-based event rates by treatment group (figure 3). No differentiation present between the CRT and control groups for subgroup specified event numbers.
5	REVERSE	Linde C, Abraham WT, Gold MR, et al. Randomized trial of cardiac resynchronization in mildly symptomatic heart failure patients and in asymptomatic patients with left ventricular dysfunction and previous heart failure symptoms. J Am Coll Cardiol. 2008;52(23):1834-1843. doi:10.1016/j.jacc.2008.08.027	No outcome event rates reported for sex, age, or race. Absence of meta regression precluded extraction of the odds ratios reported for sex and age in figure 4 (see study limitations).

*Extraction occurred from secondary analyses of the original trial, but no data was extracted from the original trial itself. **Abbreviations:** CRT = cardiac resynchronization therapy; RCTs = randomized controlled trials.

Supplemental Table S3: Excluded CRT RCTs Secondary Analyses During Data Extraction.

Study #	Trial Name	Study Reference	Reason for Exclusion
1	CARE-HF (secondary analysis)	Cleland JG, Daubert JC, Erdmann E, et al. Longer-term effects of cardiac resynchronization therapy on mortality in heart failure [the Cardiac Resynchronization-Heart Failure (CARE-HF) trial extension phase]. Eur Heart J. 2006;27(16):1928-1932. doi:10.1093/eurheartj/ehl099	Unable to distinguish sex and age-based event rates by treatment group (figure 3).
2	CARE-HF (secondary analysis)	Cleland JG, Freemantle N, Erdmann E, Gras D, Kappenberger L, Tavazzi L, Daubert JC. Long-term mortality with cardiac resynchronization therapy in the Cardiac Resynchronization-Heart Failure (CARE-HF) trial. European journal of heart failure. 2012 Jun;14(6):628-34.	Non-uniform age ranges not amenable to data extraction by ≥65 or <65 years.
3	CARE-HF (secondary analysis)	Uretsky BF, Thygesen K, Daubert JC, Erdmann E, Freemantle N, Gras D, Kappenberger L, Tavazzi L, Cleland JG. Predictors of mortality from pump failure and sudden cardiac death in patients with systolic heart failure and left ventricular dyssynchrony: results of the CARE-HF trial. Journal of cardiac failure. 2008 Oct 1;14(8):670-5.	Inadequate study duration due to short follow-up time precluding derivation of one year event rates.
4	MADIT-CRT (secondary analysis)	Biton Y, Kutyifa V, Zareba W, et al. Long-term outcome with cardiac resynchronization therapy in mild heart failure patients with left bundle branch block from US and Europe MADIT-CRT. Heart Fail Rev. 2015;20(5):535-543. doi:10.1007/s10741-015-9499-2	Event numbers for sex and age differentiated by center of enrollment rather than by CRT/control group
5	MADIT-CRT (secondary analysis)	Goldenberg I, Kutyifa V, Klein HU, et al. Survival with cardiac-resynchronization therapy in mild heart failure. N Engl J Med. 2014;370(18):1694-1701. doi:10.1056/NEJMoa1401426	Unable to distinguish sex and age-based event rates by treatment group (figure 3).
6	MADIT-CRT (secondary analysis)	Thomas S, Moss AJ, Zareba W, McNitt S, Barsheshet A, Klein H, Goldenberg I, Huang DT, Biton Y, Kutyifa V. Cardiac resynchronization in different age groups: a MADIT-CRT long-term follow-up substudy. Journal of cardiac failure. 2016 Feb 1;22(2):143-9.	Non-uniform age ranges not amenable to data extraction by ≥65 or <65 years.
7	REVERSE (secondary analysis)	Daubert C, Gold MR, Abraham WT, et al. Prevention of disease progression by cardiac resynchronization therapy in	No outcome event rates reported for sex, age, or race.

		patients with asymptomatic or mildly symptomatic left ventricular dysfunction: insights from the European cohort of the REVERSE (Resynchronization Reverses Remodeling in Systolic Left Ventricular Dysfunction) trial. J Am Coll Cardiol. 2009;54(20):1837-1846. doi:10.1016/j.jacc.2009.08.011	
8	REVERSE (secondary analysis)	Linde C, Abraham WT, Gold MR, Daubert C; REVERSE Study Group. Cardiac resynchronization therapy in asymptomatic or mildly symptomatic heart failure patients in relation to etiology: results from the REVERSE (Resynchronization Reverses Remodeling in Systolic Left Ventricular Dysfunction) study. J Am Coll Cardiol. 2010;56(22):1826-1831. doi:10.1016/j.jacc.2010.05.055	No outcome event rates reported for sex, age, or race.

Abbreviations: CRT = cardiac resynchronization therapy; RCTs = randomized controlled trials.

Supplemental Table S4. Study Details for Trial Leads, Sites and Authorship of Included CRT RCTs. [a]

Parameter	Frequency (%)	Parameter	Frequency (%)
Trial Origin		**Trial Enrollment sites**	
North America	4 (67%)	North America	5 (29%)
Europe	2 (33%)	Europe	6 (35%)
Australia	0	Australia	3 (18%)
Asia	0	Asia	3 (18%)
Africa	0	Africa	0
South America	0	South America	0
First author by sex		**Corresponding author by sex** [a]	
Male	4 (67%)	Male	5 (71%)
Female	2 (33%)	Female	2 (29%)
First author by race		**Corresponding author by race** [a]	
White	4 (67%)	White	5 (71%)
Black	0	Black	0
Hispanic	0	Hispanic	0
Asian	2 (33%)	Asian	2 (29%)
Southeast Asian	0	Southeast Asian	0
Other	0	Other	0
First author by site		**Corresponding author by site** [a]	
North America	3 (50%)	North America	4 (57%)
Europe	3 (50%)	Europe	3 (43%)
Australia	0	Australia	0
Asia	0	Asia	0
Africa	0	Africa	0
South America	0	South America	0
Funding recipient by sex [b]		**Funding sponsor by trial**	
Male	5 (71%)	Industry	4 (67%)
Female	2 (29%)	Peer-reviewed	2 (33%)
Funding recipient by race [b]		**Funding recipient by site** [b]	
White	5 (71%)	North America	4 (57%)
Black	0	Europe	3 (43%)
Hispanic	0	Australia	0
Asian	2 (29%)	Asia	0
Southeast Asian	0	Africa	0
Other	0	South America	0

[a] Biton et al. (2015) reported two corresponding authors. [b] Funding recipient, if not explicitly reported, was defaulted to the listed corresponding author based on the manuscript or from the clinicaltrials.gov website.
Abbreviations: CRT = cardiac resynchronization therapy; RCTs = randomized controlled trials.